

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

Student Name: Vijayasri

Roll No.: A24126552262

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1. TITLE

Develop a Mini Student Profile Generator Using HTML, CSS and JavaScript

2. AIM

To develop a mini web application using HTML, CSS and JavaScript that accepts student information through a form and dynamically displays the submitted details as a student profile.

3. OBJECTIVE

The objectives of this experiment are:

- To create a simple student profile form using HTML.
 - To style the application using CSS.
 - To use a JavaScript class for storing student information.
 - To read values entered in the form fields.
 - To create and display the student profile dynamically.
 - To understand DOM manipulation and event handling.
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4. THEORY

HTML is used to structure the web page and provide the input fields for student information. **CSS** is used to control the layout, spacing, fonts, colors and appearance of the application. **JavaScript** is used to process the entered values and dynamically create the student profile.

In this application, a JavaScript **Student** class stores the name, roll number, department and CGPA. When the user clicks the Create Student Profile button, an event listener reads the input values, creates a Student object and generates the profile using DOM elements.

The basic flow of the application is:

User Input → JavaScript Event → Student Object → DOM Manipulation → Student Profile

5. ALGORITHM / PROCEDURE

1. Create an HTML page for the Student Profile Generator.
 2. Add input fields for Name, Roll Number, Department and CGPA.
 3. Add a button to create the student profile.
 4. Link the CSS file to the HTML page.
 5. Link the JavaScript file to the HTML page.
 6. Create a Student class with name, roll number, department and CGPA properties.
 7. Select the button and profile container using document.getElementById().
 8. Add a click event listener to the Create Student Profile button.
 9. Read the values from all input fields.
 10. Create a Student object using the entered values.
 11. Clear any previously generated profile.
 12. Create a profile div and required paragraph elements dynamically.
 13. Add the student details to the generated elements.
 14. Append the profile to the webpage.
 15. Verify the output with different student details.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Profile</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
<h1>Student Profile Generator</h1>
<div class="form">
<label for="name">Name</label>
<input type="text" id="name" placeholder="Enter your name">
<label for="rollNumber">Roll Number</label>
<input type="text" id="rollNumber" placeholder="Enter roll number">
<label for="department">Department</label>
<input type="text" id="department" placeholder="Enter department">
<label for="cgpa">CGPA</label>
<input type="number" id="cgpa" step="0.01" placeholder="Enter CGPA">
<button id="createProfile">Create Student Profile</button>
</div>
<div id="studentProfile"></div>
</div>
<script src="script.js"></script>
</body>
</html>
```

script.js

```
// Create Student class
class Student {
  constructor(name, rollNumber, department, cgpa) {
    this.name = name;
    this.rollNumber = rollNumber;
    this.department = department;
    this.cgpa = cgpa;
  }
}

// Select the button and profile container
const createProfileButton = document.getElementById("createProfile");
const studentProfile = document.getElementById("studentProfile");

// Add event listener to the button
createProfileButton.addEventListener("click", function () {
  // Get values from input fields
  const name = document.getElementById("name").value;
  const rollNumber = document.getElementById("rollNumber").value;
  const department = document.getElementById("department").value;
  const cgpa = document.getElementById("cgpa").value;

  // Create Student object
  const student = new Student(
    name,
    rollNumber,
    department,
    cgpa
  );

  // Clear previous profile
  studentProfile.innerHTML = "";

  // Create profile div dynamically
  const profile = document.createElement("div");
  profile.className = "profile";

  // Create heading
  const heading = document.createElement("h2");
  heading.textContent = "Student Profile";

  // Create paragraphs
  const namePara = document.createElement("p");
  namePara.textContent = "Name : " + student.name;

  const rollPara = document.createElement("p");
  rollPara.textContent = "Roll No : " + student.rollNumber;

  const departmentPara = document.createElement("p");
  departmentPara.textContent = "Department : " + student.department;

  const cgpaPara = document.createElement("p");
  cgpaPara.textContent = "CGPA : " + student.cgpa;

  // Add elements to profile
  profile.appendChild(heading);
  profile.appendChild(namePara);
  profile.appendChild(rollPara);
  profile.appendChild(departmentPara);
  profile.appendChild(cgpaPara);

  // Add profile to webpage
  studentProfile.appendChild(profile);
});
```

style.css

```
body {
font-family: Arial, sans-serif;
background-color: #f2f2f2;
margin: 0;
padding: 40px;
}

.container {
width: 450px;
margin: auto;
background-color: white;
padding: 30px;
border-radius: 10px;
box-shadow: 0 4px 10px rgba(0, 0, 0, 0.2);
}

h1 {
text-align: center;
margin-bottom: 25px;
}

.form {
display: flex;
flex-direction: column;
}

label {
margin-top: 10px;
margin-bottom: 5px;
font-weight: bold;
}

input {
padding: 10px;
border: 1px solid #ccc;
border-radius: 5px;
font-size: 15px;
}

button {
margin-top: 20px;
padding: 12px;
border: none;
border-radius: 5px;
background-color: #333;
color: white;
font-size: 16px;
cursor: pointer;
}

button:hover {
background-color: #555;
}

.profile {
margin-top: 25px;
padding: 20px;
border-radius: 8px;
background-color: #f8f8f8;
border-left: 5px solid #333;
}

.profile h2 {
margin-top: 0;
}
```

```
.profile p {  
font-size: 16px;  
margin: 10px 0;  
}
```

7. CODE EXPLANATION

Code	Explanation
class Student	Defines a Student class to store student information.
constructor(...)	Initializes name, roll number, department and CGPA.
getElementById()	Selects HTML elements from the page.
addEventListener("click", ...)	Runs the profile-generation code when the button is clicked.
new Student(...)	Creates a Student object using the entered values.
createElement("div")	Creates the profile container dynamically.
textContent	Adds student information to the generated elements.
appendChild()	Adds the generated elements to the profile and webpage.

8. TEST CASES AND EXECUTION DATA

The application was checked with different combinations of student information and the generated profile was observed.

Test Case	Name	Roll No.	Department	CGPA	Expected Result	Actual Result	Status
TC-01	Vijayasri	A24126552262	CSE (AI & ML)	8.65	Student profile displayed correctly	Student profile displayed correctly	Pass
TC-02	Rahul	101	CSE	8.20	Student profile displayed correctly	Student profile displayed correctly	Pass
TC-03	Priya	102	CSE	9.10	Student profile displayed correctly	Student profile displayed correctly	Pass
TC-04	Arjun	103	CSE	7.50	Student profile displayed correctly	Student profile displayed correctly	Pass

Additional Validation Test

Test Case	Input	Expected Result	Status
TC-05	Fields entered with different values	The generated profile should show the corresponding values	Pass

9. OUTPUT

Output 1: Student Profile Generator

Student Profile Generator

C:/Users/dadhi/OneDrive/Desktop/Student-Profile-JavaScript/index.html

Student Profile Generator

Name

Roll Number

Department

CGPA

Create Student Profile

Student Profile
Name : Vijayasri
Roll Number : A24126552262
Department : CSE (AI & ML)
CGPA : 8.65

Output 2: Student Input Form

Student Profile C:/Users/dadhi/OneDrive/Desktop/Student-Profile-JavaScript/index.html

Student Profile Generator

Name

Vijayasri

Roll Number

A24126552262

Department

CSE (AI & ML)

CGPA

8.65

Create Student Profile

Output 3: Generated Student Profile

Create Student Profile

Student Profile

Name : Vijayasri

Roll Number : A24126552262

Department : CSE (AI & ML)

CGPA : 8.65

10. RESULT

Thus, a mini Student Profile Generator was successfully developed using HTML, CSS and JavaScript. The application accepts student information, creates a Student object, dynamically generates the profile and displays the entered details correctly on the webpage.
